Problem Statement No. 12

Agentic AI for Personalized Course Pathways

The Challenge

Students often struggle to identify the right learning path that aligns with their interests and long-term goals due to the overwhelming number of online courses and a lack of personalized guidance. LearnMate aims to solve this by acting as an Agentic AI coach that interacts with students, understands their interests (like Frontend Development, Cybersecurity, UI/UX Design, Distributed Systems, etc.), assesses their current skill level, and dynamically builds a personalized course roadmap that adapts over time based on progress and preferences.

Key Objectives & Agentic Requirements

1. Multi-Step Learner Calibration: Capture baseline skills, experience tier, and time commitment.
2. Skill Deficit Auditing: Automated gap analysis comparing user skills against enterprise industry roles.
3. Dynamic Milestone Roadmap: Milestone DAG synthesis with progressive learning stages.
4. Environmental Feedback Adaptation: Continuous re-planning of upcoming milestones based on progress.
5. Socratic Conversational Companion: Real-time conceptual debugging grounded in technical resources.
6. Cognitive Retention & Progress Tracking: Continuous retention decay tracking with spaced repetition.

Mandatory Technology Stack

- Foundation Models: IBM Granite (granite-3-8b-instruct / ibm/granite-13b-chat-v2)
- AI Cloud Platform: IBM Cloud Lite services & IBM watsonx.ai
- Workflow Orchestration: Langflow agentic visual pipeline
- Application Architecture: Full-stack implementation (FastAPI backend + Modern React frontend)
- Evaluation Mode: Offline resilience with dual-mode simulation for zero-quota environments

Submission & Evaluation Deliverables

- Public Source Code Repository: Complete codebase with unit tests and documentation.
- Live Interactive Demonstration: Deployed web platform accessible to evaluators.
- System Architecture & Presentation: Structured overview of multi-agent workflows.
- Project Manifest (app.json): IBM Cloud / SkillsBuild standard configuration file.